

Evaluating the Interplay of Information Status and Information Content in a Multilingual Parallel Corpus

SIGTYP 2026

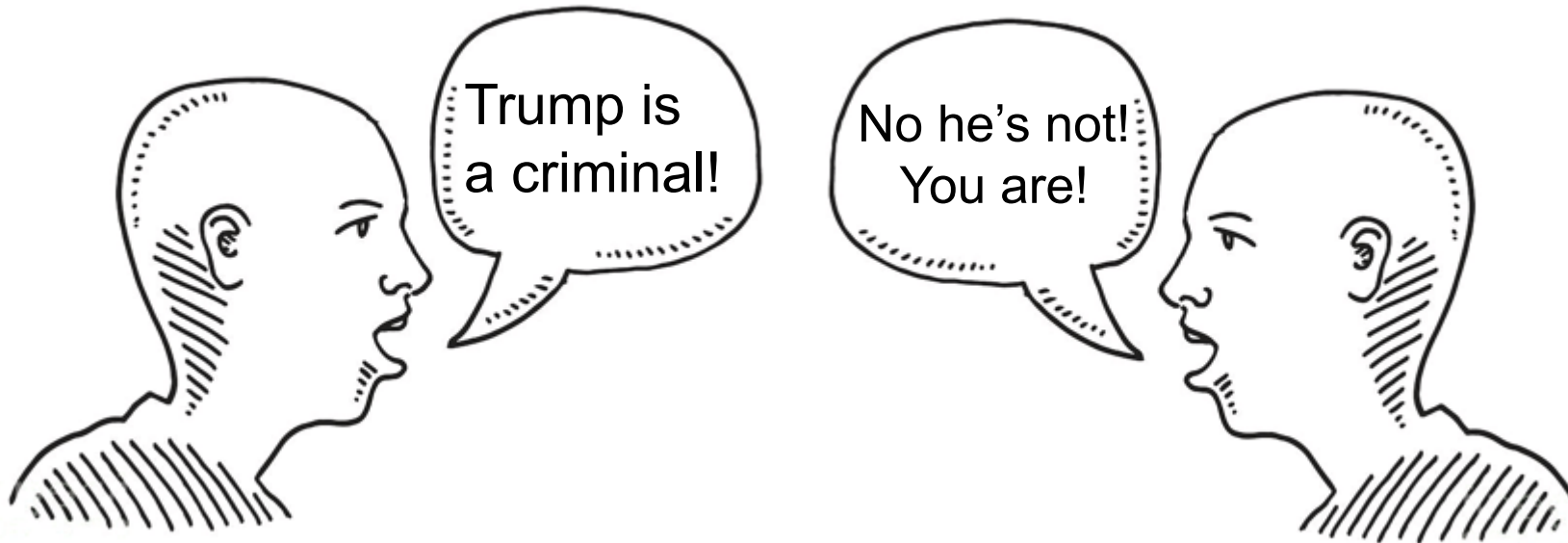
Toshiki Nakai & Julius Steuer

29.03.2026



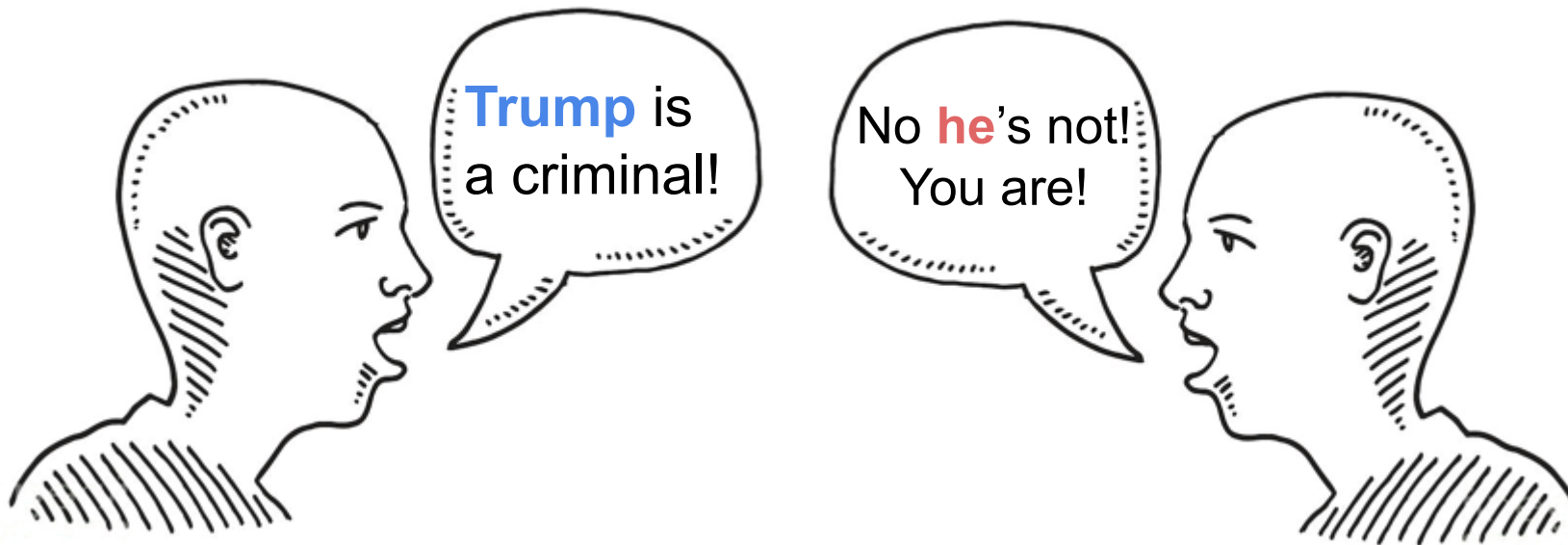
Aim: bring together various dimensions of information management

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- **Unknown/new** information is introduced one way, and **known/old** information is encoded in special ways (e.g., pronouns)

Information status & word order

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(Clark and Clark 1978)

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In a **hole** in the **ground** there lived a **hobbit**.

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In a **hole** in the **ground** there lived a **hobbit**.

It was a **hobbit-hole**, and **that** means **comfort**.

Surprisal theory (Halle 2001, Levy 2008)

- Information content (in bits) of a word given context

$$\text{surprisal}(\text{word}|\text{context}) \propto -\log p(\text{word}|\text{context})$$

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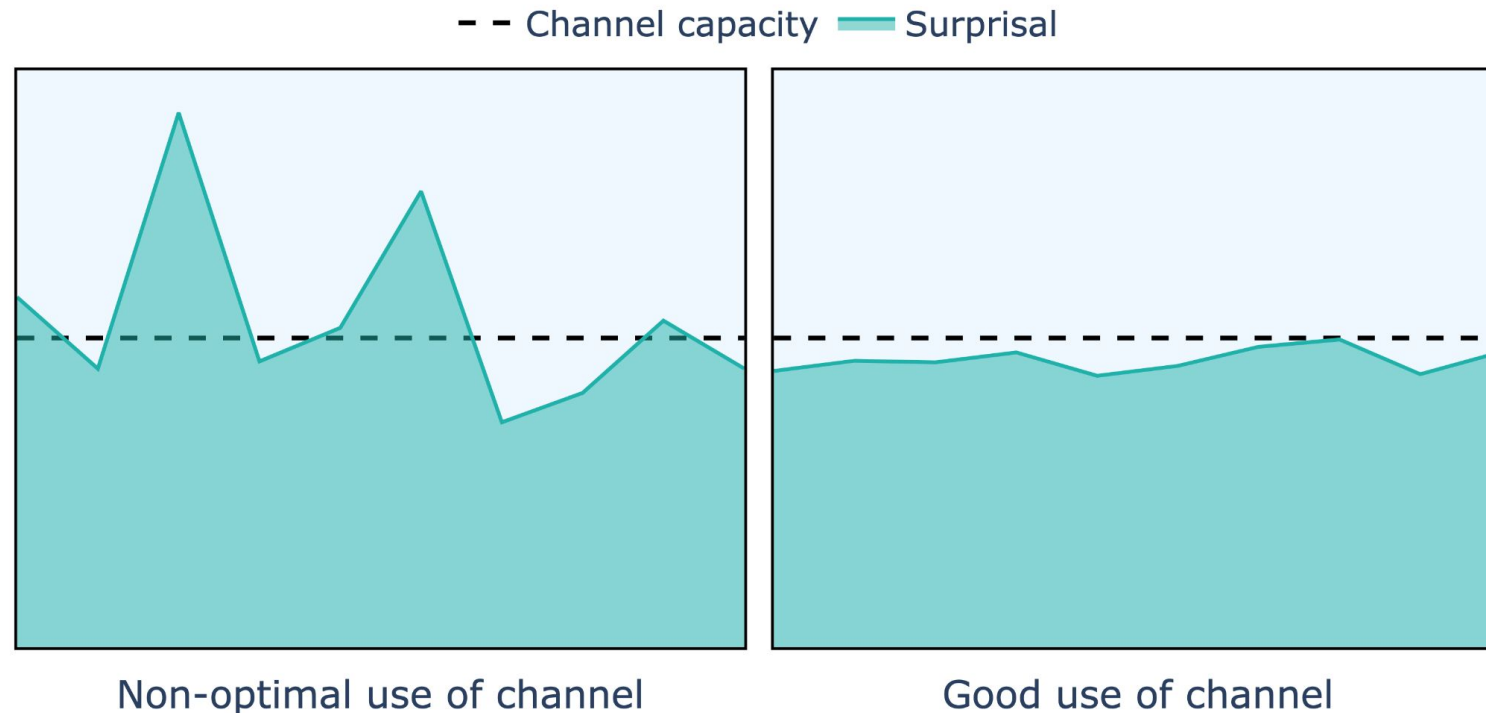
- Information content (in bits) of a word given context

$$\text{surprisal}(\text{word}|\text{context}) \propto -\log p(\text{word}|\text{context})$$

- Surprisal correlates with measures of human processing effort, e. g.
 - Reading times
 - N400/P600 EEG signals

Uniform information density (UID)

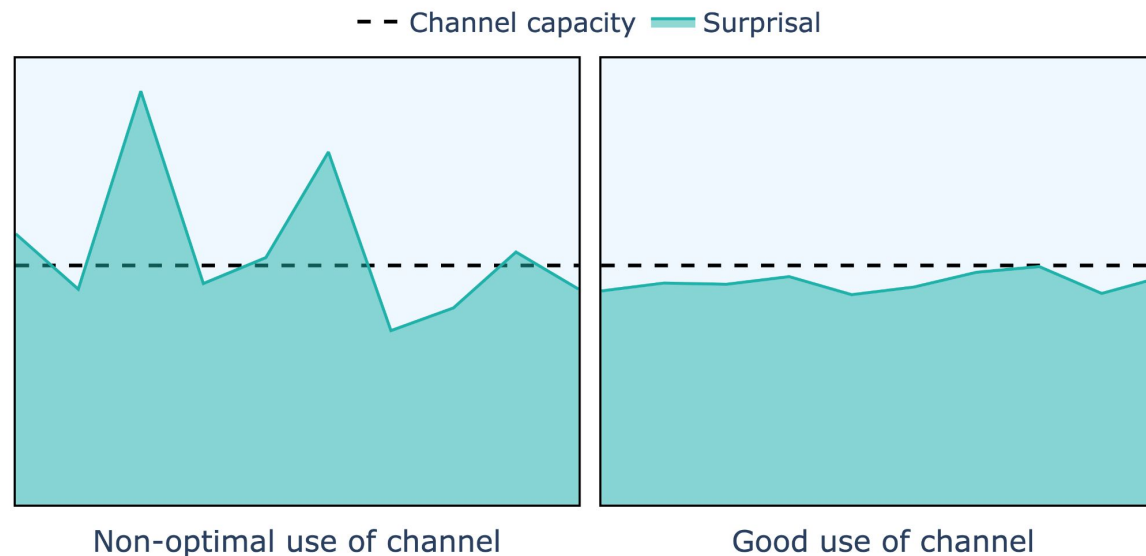
- Speakers prefer information **to be spread equally** throughout the sentence (Levy and Jaeger 2007)



Adapted from Crocker et al. (2016)

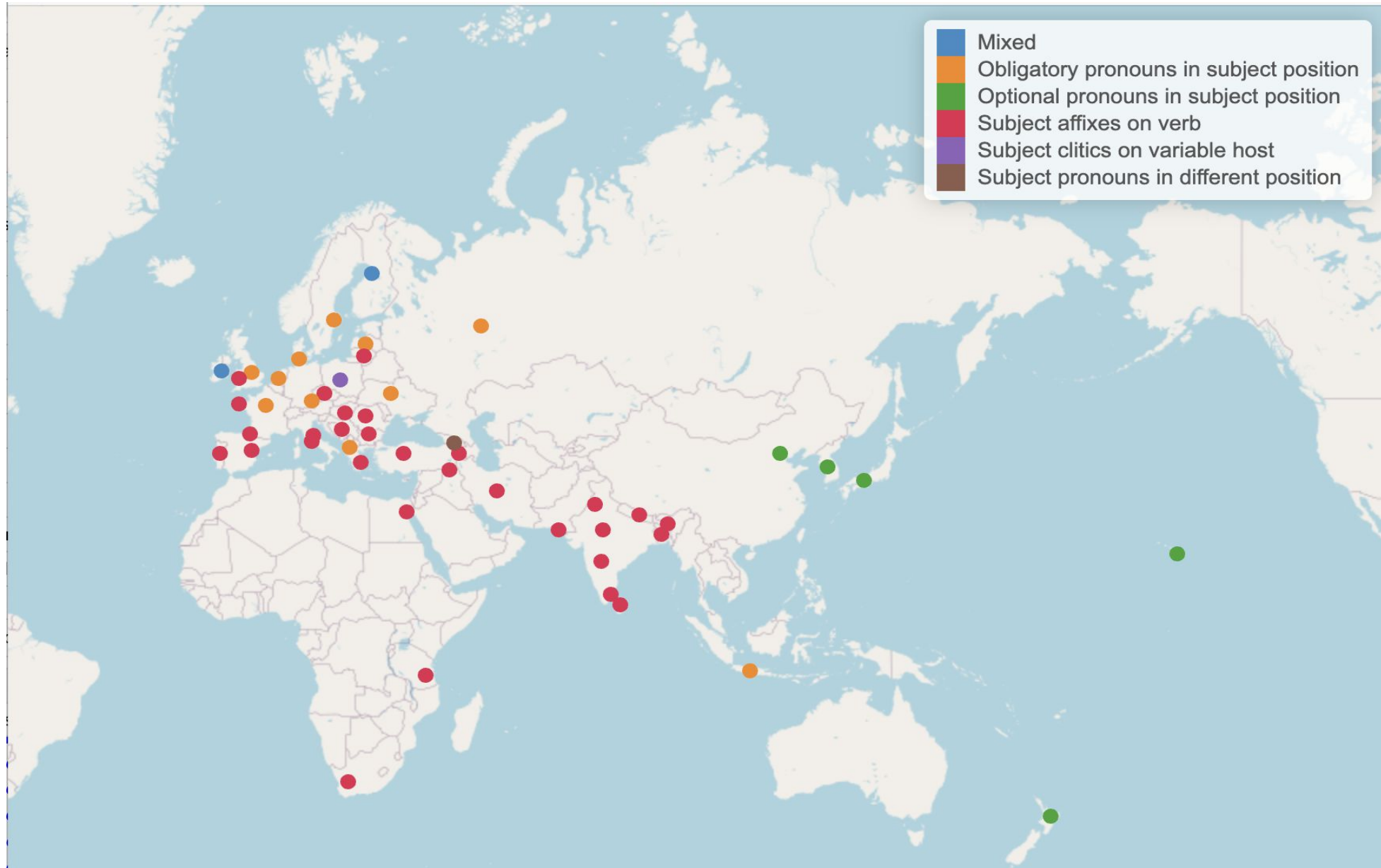
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- Does **information status** (given vs. new) align with **information density**?

Sample (according to Dryer 2013)



Workflow

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 - a. We have some gold-level-annotation for Cieplnf in other languages to test the success of projection (**experiment A**)
 - b. Not all languages use overt NPs/pronouns as frequently as English does; **experiment B** on overt reference using gold-aligned data

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1. **CieplInf English**: Information status annotations on English mini-CIEP+
2. **Annotation project** to all sample languages – two caveats:
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3. Assess the **correlation** between given/new status and information density using a variety of methods

CieplInf

A Multilingual Parallel Corpus for Coreference Resolution and Information Status in the Literary Domain

Andrew Dyer, Ruveyda Betul Bahceci, Maryam Rajestari,
Andreas Rouvalis, Aarushi Singhal, Yuliya Stodolinska,
Syahidah Asma Umniyati, Helena Rodrigues Menezes de Oliveira Vaz

- Manually annotated for information status (Dyer et al. 2024)
- *New value*: applied to the **first mention** of an entity or to **another mention** that expands the **known information** about the entity

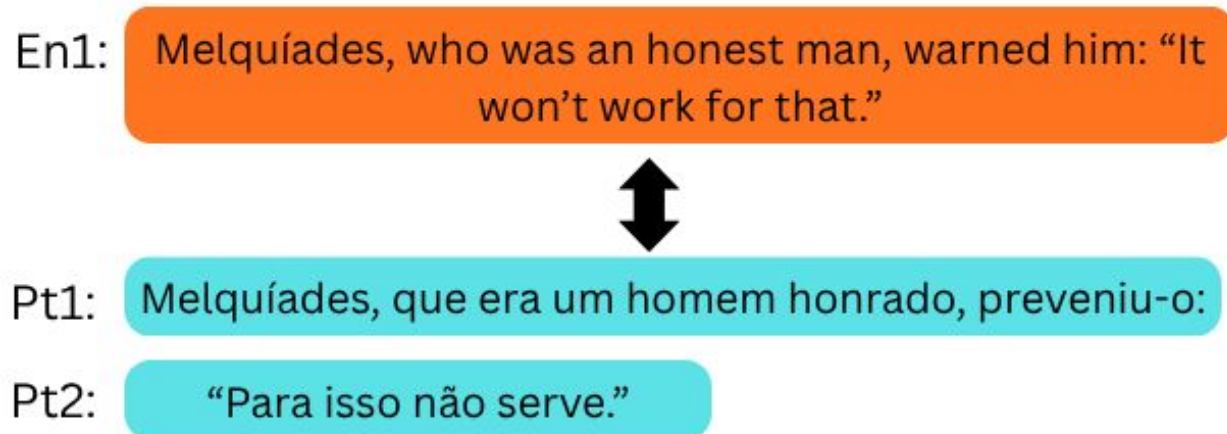
Attribute	Values	Required
InfStat	new, given	true
CorefType	ana, cata, pred, disc, appos, coref	true
Indexing	NullSubj, NullObj, NullPoss	false
Bridging	<i>boolean</i>	false
Deixis	<i>boolean</i>	false

Table 1: The set of attributes that we can apply to a mention.

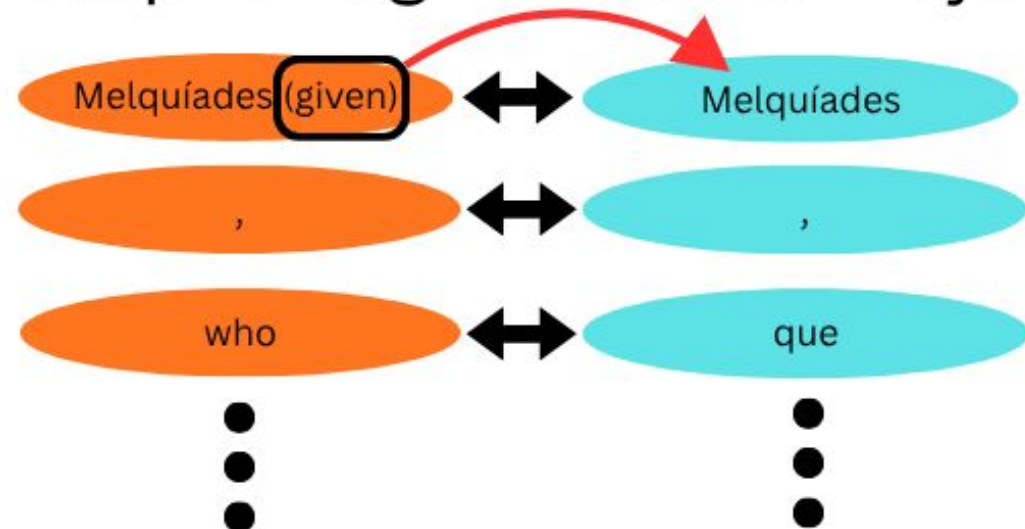
Information status projection

- Sentence alignment: Bertalign (Liu & Yhu 2022)
- Word alignment: Awsome Align (Dou & Neubig 2021)

Step 1: Align Sentences



Step 2: Align Words & Project



En: **Melquíades**, who was an honest **man**, warned **him**: “It won’t work for **that**.”

Fr: **Melquiades**, qui était un **homme** honnête, **le** mit en garde : « Ça ne sert pas à ça. »

Melquiades InfStat=given|alignment=Melquíades

homme InfStat=given|alignment=man

le InfStat=given|alignment=him

Zh: **梅尔基亚德斯** 是个诚实的 **人**，当时就提醒 **他**：“干不了 **这个**。”

梅尔基亚德斯 InfStat=given|alignment=Melquíades

人 InfStat=given|alignment=man

他 InfStat=given|alignment=him

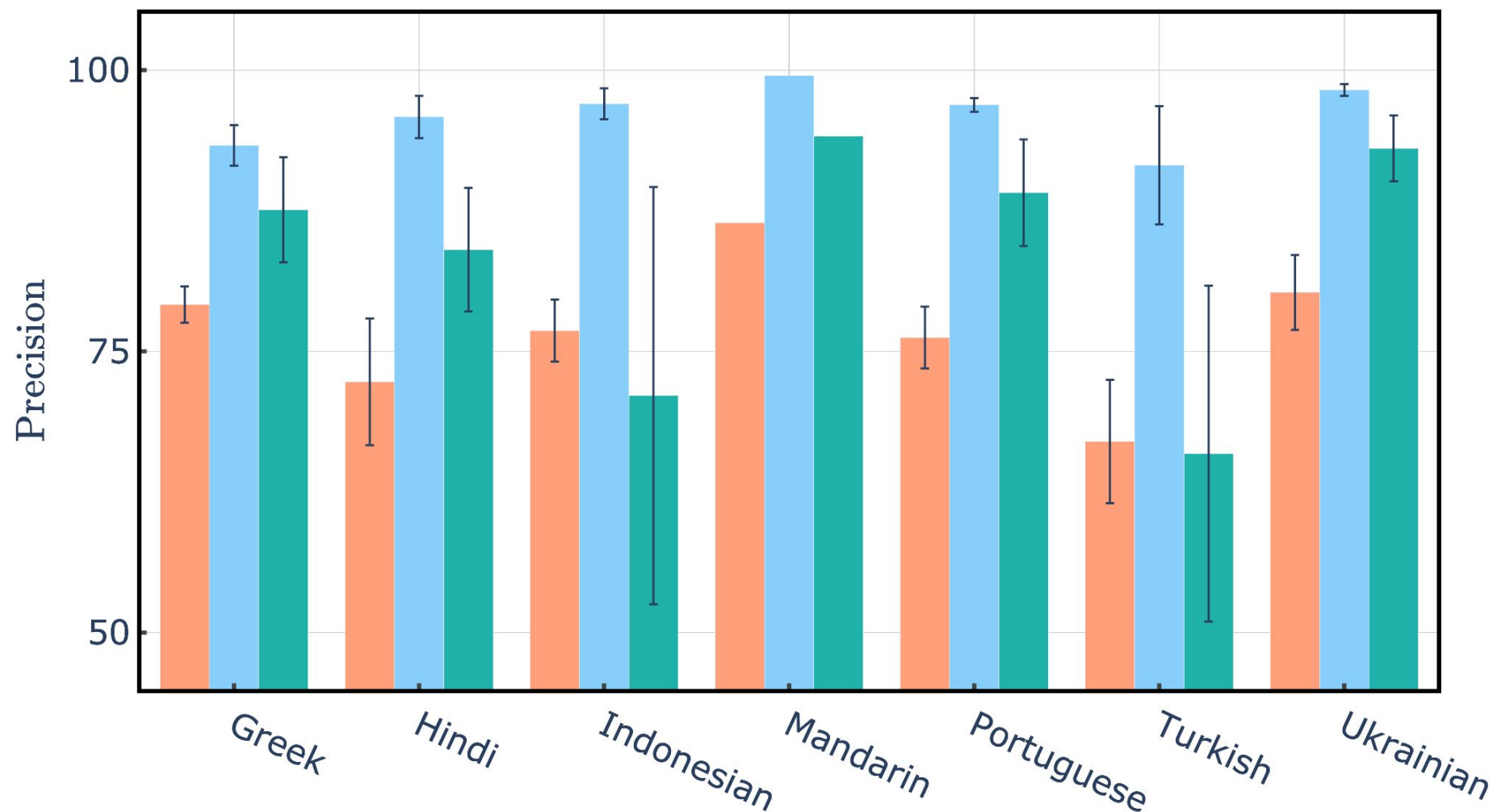
这 InfStat=given|alignment=that

Projection results (gold data)

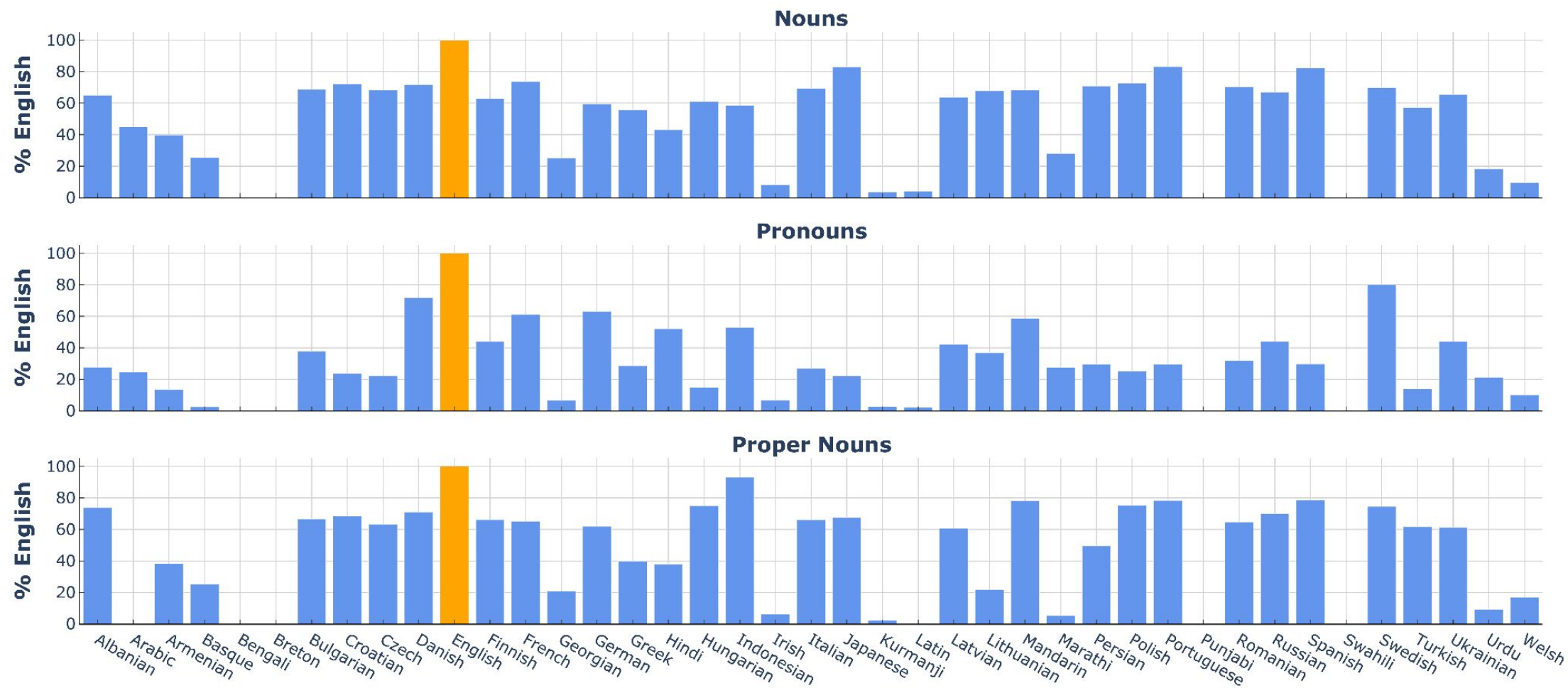
Nouns

Pronouns

Proper Nouns



Projection on unlabeled corpus



Annotation projection: Outlook

- **NOUNs:**

- Precision is lower for non-SVO languages (Hindi, Turkish)
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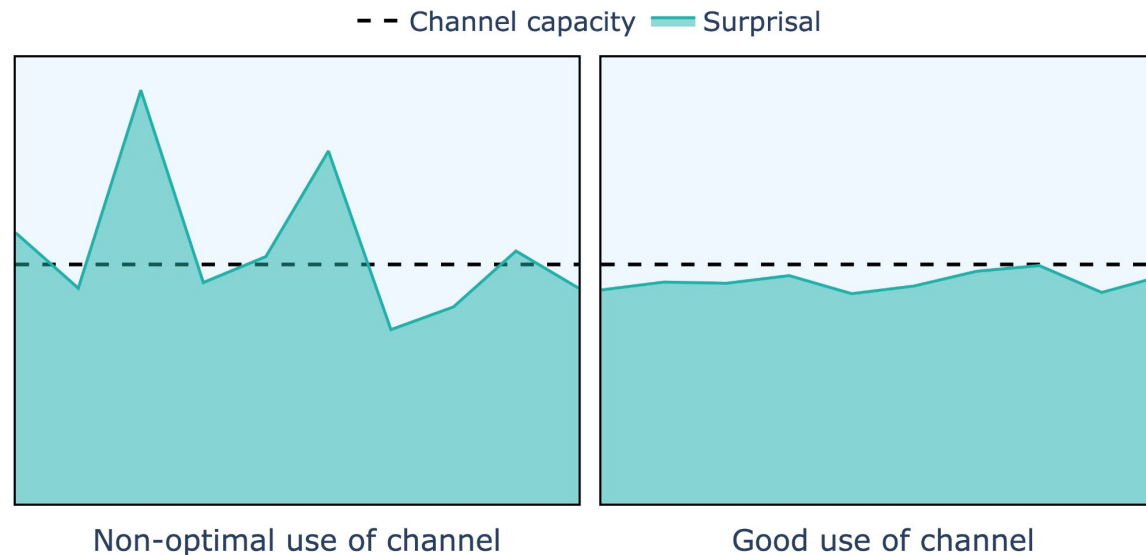
- Projection is more accurate because they are easier to align (high frequency, and mostly “given”)
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- **PROPNS:**

- Precision varies depending on the language.
- Language-specific usage pattern?

Uniform information density (UID)

- Speakers prefer information **to be spread equally** throughout the sentence (Levy and Jaeger 2007)



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Information status & UID

- Based on how much information there is in the context of a word:

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$$\text{surprisal}(\text{given}) < \text{surprisal}(\text{new})$$

- If UID holds, we expect linguistic *form* (word order, lexical choice, construction type, ...) to accommodate (potentially) high/low surprisal items such that:

$$\text{surprisal}(\text{given}) \approx \text{surprisal}(\text{new})$$

Methodology

- Estimate surprisal on our projected corpus with a multilingual language model (mGPT, Shliazhko et al. 2024)
- Calculate per-language deviation $\text{UID}_{\mathcal{L}}$ from channel capacity $\hat{S}_{\mathcal{L}}$ for **given** and **new** items:

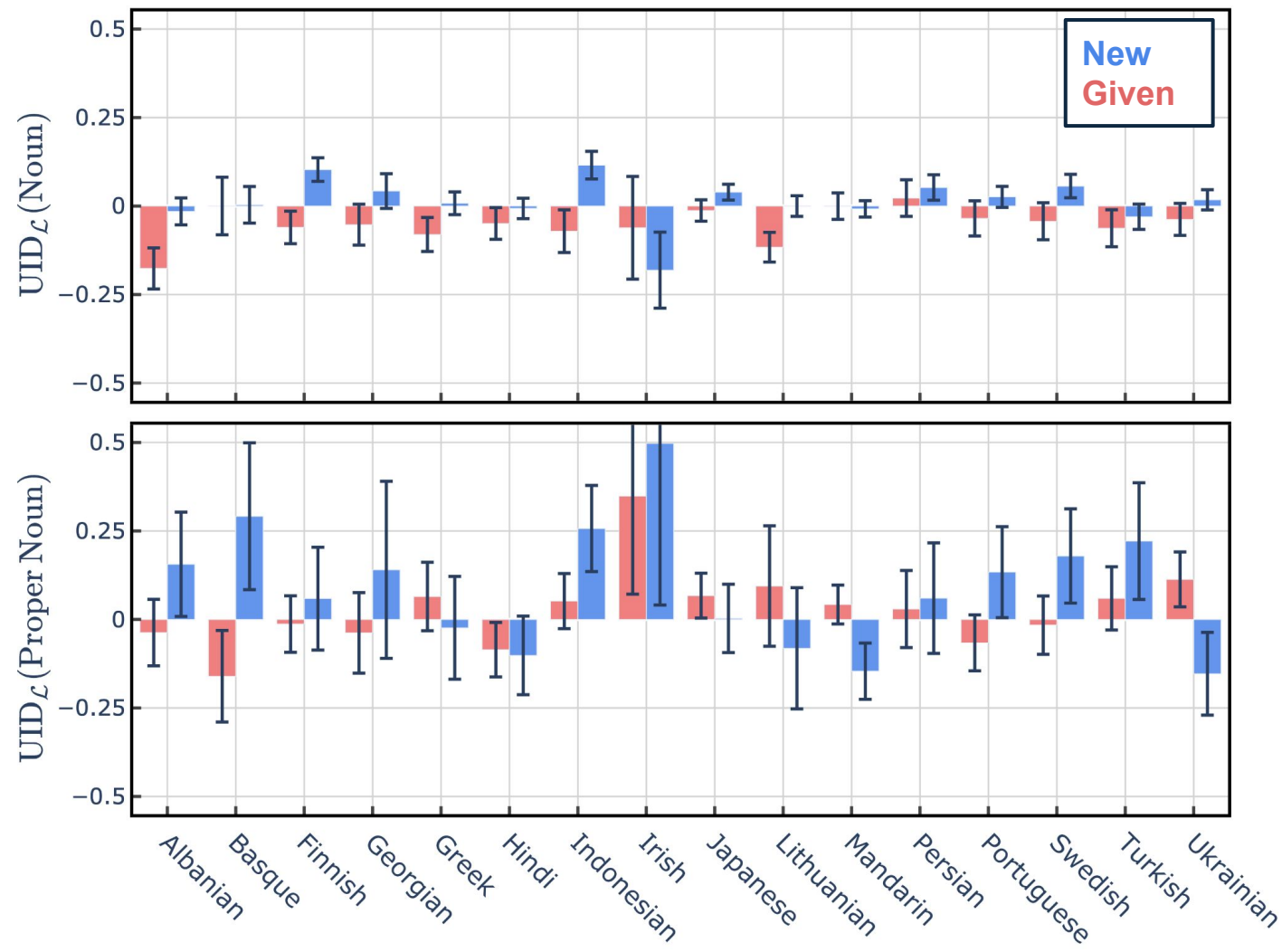
$$\text{UID}_{\mathcal{L}} = \frac{1}{N-1} \sum_{i=2}^N (S(\mathbf{s}_i | \mathbf{c}) - \hat{S}_{\mathcal{L}})^2$$

Research Question*

Does $\text{UID}_{\mathcal{L}}$ hold for given and new items, i.e., is there a difference in surprisal between new and given items when viewing deviation from channel capacity?

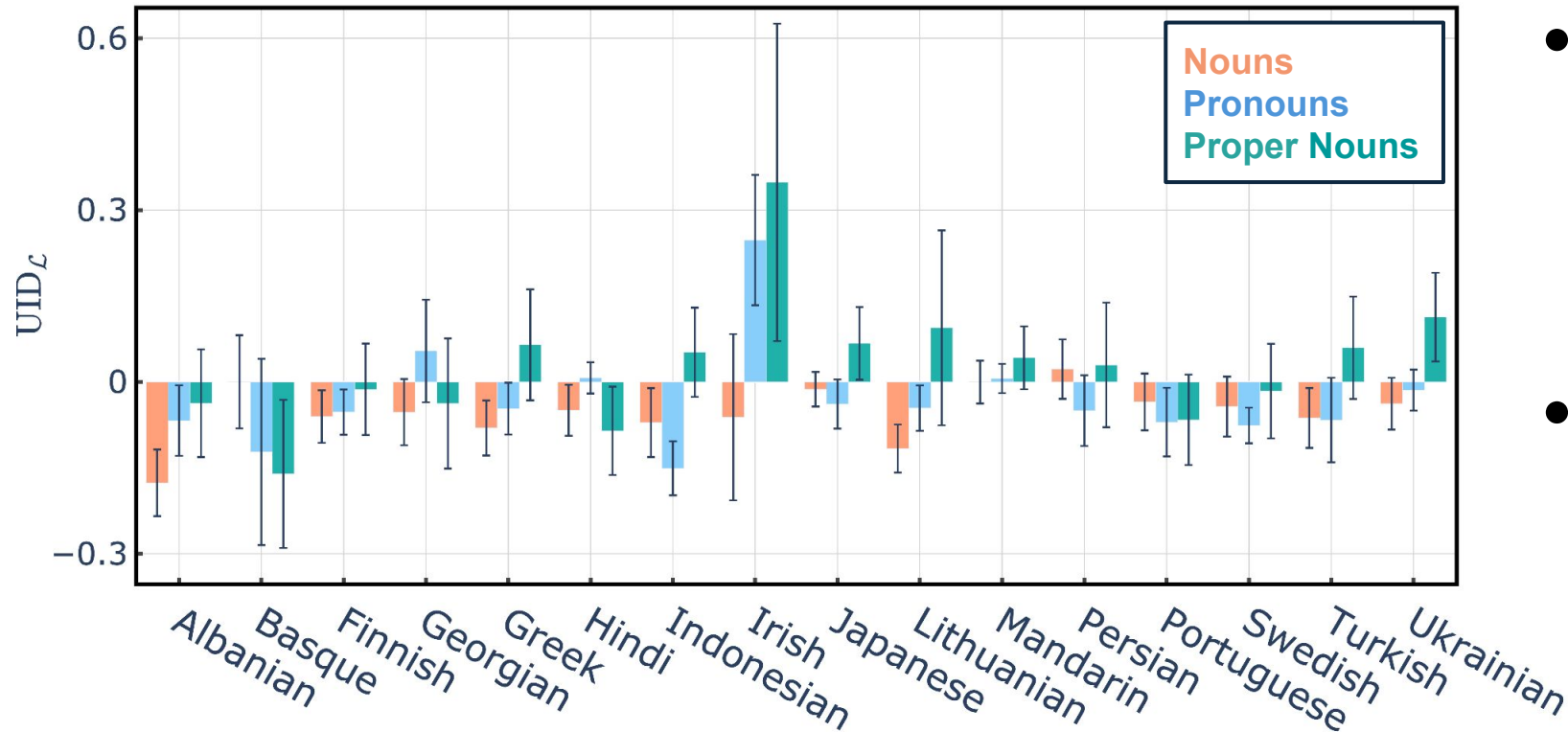
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Preliminary results



- Surprisal of nouns stays close to channel capacity independently of IS
- Effect of IS on the surprisal of proper nouns
- Nouns more conformant to UID than proper nouns?

Preliminary results: Givenness



- Surprisal of given nouns, pronouns usually falls below channel capacity
- Surprisal of proper nouns exceeds it for some languages
- Generally small effect sizes

Conclusion & future directions

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- CIEPInf annotation projection affected by quality of alignment models
- Analysis limited by coarse-grained annotation (**new** vs. **old**)
- **Mediated** information status (Nissim et al. 2004)

It's hard to raise **one child** without **them** thinking **they're** the pivot point of the universe.

The **Earth** revolves around the **Sun**, not around **you**.